

Bethe–Salpeter Campaign Equations and the r_e/r_0 Story

An author-review packet for Tepper Gill

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Generated as an author-review packet, not a journal submission.

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For author review

To: Tepper Gill (senior author, DRQM I)

From: Trey Morris

Subject: Bethe–Salpeter campaign + r_e/r_0 story — packet for your fine-tooth-comb review

This document is an **author-review packet, not a journal submission**. Findings concerning DRQM I are surfaced here “for author review” (in the convention of `CLAUDE.md` § *What this repo is*), not as “corrections.” The packet consolidates the campaign’s 28 results, the $Z=1$ empirical triangulation of r_e/r_0 , and the multi- Z scan that brings the Branch A vs. Branch C question to a head.

The packet is organised around two questions you are the only one qualified to answer:

1. Does DRQM I §III.D leave the cutoff free per state (the reading PR #87 takes), or does it fix a Z -universal cutoff (the reading PRs #84/#85/#86 take)?
2. Does the framework’s intended position match *hypothesis (ii)* (the proper-time one-loop vertex correction reproduces standard-QED Schwinger by inheritance, with all dual structure absorbed into the matter-side kernel), or *hypothesis (i)* (an independent proper-time vertex computation that produces $a_e^{(1)} = \alpha/(2\pi)$ from the framework’s apparatus)?

The Mathematica notebooks are translated to LaTeX equation blocks in Equations. The campaign-closing honest-scope statement (“zero of 28 results constitute independent corroboration distinct from textbook QM,” per `10_CrossComparison.md` §1) is preserved verbatim. The numbered closing-question list (§10) is the place to leave your verdict.

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Abstract

The Bethe–Salpeter precision-predictions campaign rederives 28 canonical results in atomic and few-electron physics under the proper-time / dual-theory formulation of DRQM I, comparing each prediction against CODATA-2018 / PDG-2024 measured values. Of the 28 results, 20 are formulation-independent (the framework agrees with textbook because it reduces to textbook QM where the observable lives), six are g_s -dependent observables back-fit to measured g_s at the $Z=1$ triangulated r_e/r_0 , and two are at Bethe-estimate precision (Lamb shift and helium ground state). The campaign’s load-bearing honest-scope statement, preserved verbatim throughout: **zero of 28 results constitute independent corroboration of the dual theory’s content distinct from textbook QM** (10_CrossComparison.md §1).

The $Z=1$ triangulated cutoff value $r_e/r_0 = 0.4994205099128317 \pm 2.5 \times 10^{-13}$ (PR #62) is the joint best-fit across the six g_s -dependent observables under the only weighting that respects the framework’s Bethe-estimate precision floor. The closed-form algebraic inversion of the (III.22) cutoff-anomaly identification yields $r_e/r_0 = (2 - a_e)/(2(2 + a_e))$, which reproduces the triangulated value to 3.45×10^{-13} when CODATA-full a_e is supplied (Dual_Relativistic_Quantum_Mechanics_I.md §III.D-extension). Finding 2’s verdict trajectory: $\bullet \rightarrow \bullet \rightarrow \bullet/\bullet$ characterised $\rightarrow \bullet/\bullet$ at hypothesis-(ii) framework precision. Unconditional $\checkmark$ requires the proper-time photon-propagator re-derivation tracked in issue #75.

A multi- Z scan across $Z = 2, 6, 8, 14, 50$ (PR #87, $^3\text{He}^+$ through $^{118}\text{Sn}^{49+}$) tests the (Z-i) universal-cutoff vs. (Z-ii) Z -scaled-cutoff readings of (III.22)/(III.23). The (Z-i) hypothesis (single r_e/r_0 fixed at the $Z=1$ triangulated value across all Z) is excluded at $\chi^2 \approx 1.0 \times 10^{16}$. The per- Z back-fit $x^{(Z)} = c_0 + c_2(Z\alpha)^2 + c_4(Z\alpha)^4$ recovers $c_0 = x_{\text{univ}}$ (the $Z=1$ triangulated value) as $Z \rightarrow 0$ to 10^{-7} , with $c_2 \approx 1/6$ matching the QED bound-state leading coefficient. The $(Z\alpha)^2$ form is a property of QED’s $g(Z\alpha)$ *inherited* by the back-fit, not derived from the framework apparatus. The Z -axis verdict is Outcome C (per- Z -specific via QED-inheritance), analogous to PR #70’s lepton-axis verdict (particle-specific via a_ℓ).

The Li^{2+} Z -extension (PRs #84/#85/#86) reaches “Branch A by construction” from the algebra (the cutoff formula has no Z), but three of four Li^{2+} measurements in the original brief were mis-provenanced and the comparisons are blocked. This packet presents both readings — the algebraic Branch A and the empirical Branch C from the Z -scan — without endorsement, and asks the senior author to adjudicate. The reconciliation is the focus of the closing question list (§10).

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1 Campaign overview — 28-result inventory

The Bethe–Salpeter precision-predictions campaign closes in `Roadmapping/Quantum_Mechanics/Bethe_Salpeter`. The full inventory of 28 results across PRs A–I is reproduced below verbatim from §1 of that document.

Per-PR result inventory.

PR	Section	Result	Notes
A	BS-§3	Schrödinger equation, operator identity	$K + V_0 = mc^2 + p^2/(2m) + V_0$ exactly (Wolfram-MCP verified)
A	BS-§4	Hydrogen spectrum E_n	Matches Rydberg R_∞ to all orders in non-rel
A	BS-§5	Bohr radius + radial eigenfunctions	Identical to textbook
A	BS-§6	$O(4)$ symmetry + degeneracy	Inherited from Schrödinger algebra
B	BS-§8	Dipole matrix elements	Operator + wavefunction identity
B	BS-§10	Oscillator strengths + TRK sum rule	$[r, K] = i\hbar p/m$ survives rest-energy offset
B	BS-§13	Radial integrals + selection rules	Formulation-independent
C*	BS-§14.1	Sommerfeld–Dirac spin–orbit	Dual-Dirac FW reproduces leading $(Z\alpha)^2$
C*	BS-§14.2	$2P_{3/2}$ – $2P_{1/2}$ splitting	–7 MHz residual at Bethe-estimate precision
C*	BS-§14.3	Relativistic kinetic + Darwin	Same coefficient in dual Dirac FW
D	BS-§15	Nuclear recoil μ/m_e	Two-body kinematic identity
D	BS-§17	$(Z\alpha)^4$ Pauli/FW expansion	Weyl ordering reproduces textbook
D	BS-§18	Bethe–Salpeter equation	Apparatus inherited (ladder approximation)
E*	BS-§19	Self-energy (Bethe 1947)	Matrix elements + energy denominators formulation-independent

PR	Section	Result	Notes
E*	BS-§20	Lamb shift $2S_{1/2}-2P_{1/2}$	~ 1016 MHz vs. $1057.845(9)$ measured; ~ 42 MHz residual is standard Bethe-estimate gap
E*	BS-§21	Uehling vacuum polarisation	Identical one-loop
F*	BS-§22.1	H 1S hyperfine (21-cm)	-0.4 MHz residual at Bethe-estimate precision
F*	BS-§22.2	Muonium + positronium (deferred)	(PR I)
G	BS-§24	Photoionisation K-edge	Cross-section formulation-independent
G	BS-§30	M1 + E2 multipole transitions	M1 inherits $(g_s/-2)^2 \times$ textbook via g_s
G	BS-§35	Third-term effect on dipole approximation	Doubly suppressed; below precision floor
H	BS-§47	Helium HF $Z' = 27/16$	Variational identity
H	BS-§52	Hylleraas correlation expansion	^4He spin-singlet escapes r_e
H	BS-§60	Full helium ground-state energy	~ 6 meV residual
I	BS-§64	Helium singlet-triplet exchange	Formulation-independent (Fermi statistics)
I	BS-§72	Helium 3P_J fine structure	$\sim \text{kHz}$ residual after spin-spin/spin-orbit recombination
I	BS-§80	Positronium ortho-para + muonium hyperfine	Muonium ~ 0.1 MHz residual; positronium provisional

Total: 28 results across 10 PRs. Of these:

- **20 unconditional** — but read this honestly: every one is annotated “formulation-independent,” “operator + wavefunction identity,” “identical to textbook,” “inherited from Schrödinger algebra,” “Fermi statistics,” etc. They are agreement-by-structure: the dual-theory framework’s prediction *equals* the textbook prediction because the framework reduces to textbook QM at the level the observable engages.
- **6 at the triangulated r_e** (precision g_s -dependent observables) — all six are computed as $(g_s/-2)^n \times (\text{textbook leading term})$ with $n \in \{1, 2\}$. The triangulated $r_e/r_0 = 0.4994205099128317$ substitutes $g_s = -2.00231930\dots$ (the measured value, by which the joint fit defines r_e) and gets textbook agreement by construction.
- **2 at Bethe-estimate precision** (Lamb shift + helium ground state) — the framework, where its non-relativistic-equivalent piece is exercised, reproduces the textbook Bethe-estimate calculation and inherits its residual against measurement.

Honest scope statement (load-bearing; do not soften).

Zero of 28 results constitute independent corroboration of the dual theory’s content distinct from textbook QM. The campaign’s value is in (i) tracing where the dual theory does and does not reduce to textbook, (ii) flagging the r_e finding’s downstream self-consistency at high precision, and (iii) identifying the future-work direction (proper-time one-loop dual-QED) where a *distinct* dual-theory prediction could be derived and tested.

— 10_CrossComparison.md §1

[for **Tepper**: Please confirm the per-PR result table accurately reproduces your reading of the campaign inventory, and that the load-bearing honest scope statement is faithfully transcribed.]

2 From the Dual Dirac Equation to each observable's prediction

This section transcribes the derivation chain from Roadmapping/Equation_Verification/Dual_Relativistic (DRQM I, §§II–III) and the per-PR Bethe–Salpeter section docs (01_NonRelHydrogen.md, 03_FineStructure.md, 05_LambShift.md, 06_Hyperfine.md). No new physics is introduced; the chain is presented in the form the senior author actually reads (PDF with LaTeX equations, not Mathematica notebook). Every numbered DRQM I equation from (II.1) through (III.22) that carries a typeset form in the verification doc is surfaced below for fine-grain review, in order. Equations DRQM I records structurally rather than typesetting (the intermediate (III.11)–(III.16) spherical-coordinate steps) are noted honestly as such; their three load-bearing results (III.17-equivalent) are displayed.

2.1 The proper-time canonical form

DRQM I introduces three dual relativistic wave equations, all of the form

$$i\hbar \frac{\partial \Psi}{\partial \tau} = K \Psi, \quad K = \frac{H^2}{2mc^2} + \frac{mc^2}{2}, \quad H \in \{H_{\text{Dirac}}, H_{\text{sqr}(1)}, H_{\text{sqr}(2)}\}. \quad (1)$$

The three are distinguished by which H is substituted into (1).

2.2 The Dual Dirac Equation, Eq. (II.1)

With $H_{\text{Dirac}} = c\boldsymbol{\alpha} \cdot \boldsymbol{\pi} + \beta mc^2 + V$ and $\boldsymbol{\pi} = \mathbf{p} - e\mathbf{A}/c$, the Dual Dirac Equation is

$$i\hbar \frac{\partial \Psi}{\partial \tau} = \left\{ \frac{\boldsymbol{\pi}^2}{2m} + \beta V + mc^2 - \frac{e\hbar \boldsymbol{\Sigma} \cdot \mathbf{B}}{2mc} + \frac{V \boldsymbol{\alpha} \cdot \boldsymbol{\pi}}{mc} - \frac{i\hbar \boldsymbol{\alpha} \cdot \nabla V}{2mc} + \frac{V^2}{2mc^2} \right\} \Psi. \quad (2)$$

[Wolfram-MCP: FullSimplify of $K_{\text{Dirac}} - (\text{II.1})$ returns 0 (DRQM I §II.1 verified, 2026-05-11).]

2.3 The cleanest form: Eq. (II.3)

Eq. (II.3) (DRQM I §II.3) takes $H_{\text{sqr}(2)} = \beta \sqrt{c^2 \boldsymbol{\pi}^2 - e\hbar \boldsymbol{\Sigma} \cdot \mathbf{B} + (mc^2 + \beta V)^2}$ and yields

$$i\hbar \frac{\partial \Psi}{\partial \tau} = \left\{ \frac{\boldsymbol{\pi}^2}{2m} + \beta V + mc^2 - \frac{e\hbar \boldsymbol{\Sigma} \cdot \mathbf{B}}{2mc} + \frac{V^2}{2mc^2} \right\} \Psi. \quad (3)$$

Why this is structurally clean. Unlike (II.1) and (II.2), Eq. (II.3) is scalar-only at the operator level (apart from the trivial βV term). Three commutations together ($[\beta, \boldsymbol{\pi}] = [\beta, \boldsymbol{\Sigma}] = [\beta, V] = 0$, plus $\beta^2 = 1$) collapse the radicand without leaving operator-valued cross terms. Eq. (II.3) reads exactly like a Pauli-style wave equation with a single relativistic correction $V^2/(2mc^2)$, and in the $V \rightarrow 0$ limit reduces cleanly to the Pauli equation plus a rest-energy constant. [Wolfram-MCP: (II.3) verified 2026-05-11; see §??.]

2.4 Dirac eigenvalue problem and the spinor split

Section III recasts the Dirac equation as an eigenvalue problem with $\Psi = [\psi_1, \psi_2]^t$:

$$\begin{aligned} (\lambda - V - mc^2) \psi_1 &= c(\boldsymbol{\sigma} \cdot \boldsymbol{\pi}) \psi_2, \\ (\lambda - V + mc^2) \psi_2 &= c(\boldsymbol{\sigma} \cdot \boldsymbol{\pi}) \psi_1. \end{aligned} \quad (4)$$

Solving the second for ψ_2 :

$$\psi_2 = c[\lambda - V_0 + mc^2]^{-1} (\boldsymbol{\sigma} \cdot \boldsymbol{\pi}) \psi_1 \quad (\text{III.2}). \quad (5)$$

2.5 From H_D^2 to the cutoff approximation

Writing $H_D = H_0 + V_0$ with $H_0 = c \boldsymbol{\alpha} \cdot \boldsymbol{\pi} + \beta mc^2$ and using $\{\boldsymbol{\alpha}, \beta\} = 0$ together with the Dirac identity $(\boldsymbol{\alpha} \cdot \boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 - (e\hbar/c) \boldsymbol{\Sigma} \cdot \mathbf{B}$:

$$H_D^2 = H_0^2 + V_0^2 + \{H_0, V_0\}, \quad H_0^2 = c^2 \boldsymbol{\pi}^2 - e\hbar c \boldsymbol{\Sigma} \cdot \mathbf{B} + m^2 c^4. \quad (6)$$

Divide by $2mc^2$, add $mc^2/2$, define $V := \{H_0, V_0\}/(2mc^2)$:

$$K_D = \frac{\boldsymbol{\pi}^2}{2m} + V - \frac{e\hbar \boldsymbol{\Sigma} \cdot \mathbf{B}}{2mc} + mc^2 + \frac{V_0^2}{2mc^2} \quad (\text{III.3}). \quad (7)$$

Eq. (III.4) — K_D with the shorthand V expanded. Use $[\boldsymbol{\alpha} \cdot \boldsymbol{\pi}, V_0] = -i\hbar \boldsymbol{\alpha} \cdot \nabla V_0$ and $\{\beta, V_0\} = 2\beta V_0$ to expand $V = \{H_0, V_0\}/(2mc^2)$:

$$E \Psi = \left\{ \frac{\boldsymbol{\pi}^2}{2m} + \beta V_0 + mc^2 - \frac{e\hbar \boldsymbol{\Sigma} \cdot \mathbf{B}}{2mc} + \frac{V_0 \boldsymbol{\alpha} \cdot \boldsymbol{\pi}}{mc} - \frac{i\hbar \boldsymbol{\alpha} \cdot \nabla V_0}{2mc} + \frac{V_0^2}{2mc^2} \right\} \Psi. \quad (8)$$

[Wolfram-MCP: Same expansion as (II.1) with $V \rightarrow V_0$; residual 0 (DRQM I §III.4, 2026-05-11).]

Eqs. (III.5)–(III.6) — upper/lower spinor split. Block-decompose (8) into separate ψ_1 and ψ_2 equations. The ψ_2 equation is solved as (5). Substituting into the ψ_1 equation produces a non-trivial chain-rule term because $\boldsymbol{\sigma} \cdot \boldsymbol{\pi}$ does not commute with $1/(\lambda - V_0 + mc^2)$:

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\pi}) \left[\frac{(\boldsymbol{\sigma} \cdot \boldsymbol{\pi}) \psi_1}{\lambda - V_0 + mc^2} \right] = \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 \psi_1}{\lambda - V_0 + mc^2} + \frac{-i\hbar \boldsymbol{\sigma} \cdot \nabla V_0}{(\lambda - V_0 + mc^2)^2} (\boldsymbol{\sigma} \cdot \boldsymbol{\pi}) \psi_1. \quad (9)$$

The second piece of (9) (using $\mathbf{p} = -i\hbar \nabla$, so $\boldsymbol{\sigma} \cdot \mathbf{p} V_0 = -i\hbar \boldsymbol{\sigma} \cdot \nabla V_0$) is the source of the $(V_0/m)(\boldsymbol{\sigma} \cdot \mathbf{p} V_0)(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})/(\lambda - V_0 + mc^2)^2 \psi$ term in DRQM I line 261. The result is the ψ_1 -only eigenvalue equation (III.6).

Eq. (III.7) — cutoff approximation. Approximating $\lambda - V_0 + mc^2 \approx 2mc^2 + e^2/r$ for the Coulomb potential $V_0 = -e^2/r$ (binding energy $\sim 10^{-5} mc^2$ is dropped) and using $r_0 = e^2/(mc^2)$:

$$\psi_2 \approx \frac{c(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})}{2mc^2(1 + r_0/(2r))} \psi_1. \quad (10)$$

Eq. (III.8) — approximated eigenvalue equation. Substituting (10) into the ψ_1 equation from (III.6) — i.e. replacing every occurrence of $(\lambda - V_0 + mc^2)$ in (III.6) by $2mc^2(1 + r_0/(2r))$:

$$\left\{ \frac{\boldsymbol{\pi}^2}{2m} + V_0 + \frac{V_0^2}{2mc^2} - \frac{e\hbar \boldsymbol{\sigma} \cdot \mathbf{B}}{2mc} + \frac{V_0 (\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2}{2m^2 c^2 (1 + r_0/(2r))} - \frac{V_0 (\boldsymbol{\sigma} \cdot \mathbf{p} V_0) (\boldsymbol{\sigma} \cdot \boldsymbol{\pi})/m}{[2mc^2(1 + r_0/(2r))]^2} \right\} \psi_1 = (E - mc^2) \psi_1. \quad (11)$$

This is the approximated ψ_1 -only eigenvalue equation that DRQM I (III.18)–(III.20) then collect into three identifiable Schrödinger-beyond contributions.

2.6 Eqs. (III.9)–(III.10) — Pauli identity

The general Pauli identity (Sakurai Eq. 3.2.39):

$$(\boldsymbol{\sigma} \cdot \mathbf{X})(\boldsymbol{\sigma} \cdot \mathbf{Y}) = \mathbf{X} \cdot \mathbf{Y} + i \boldsymbol{\sigma} \cdot (\mathbf{X} \times \mathbf{Y}) \quad (\text{III.9}). \quad (12)$$

Specialising to $\mathbf{X} = \mathbf{Y} = \boldsymbol{\pi}$ with $\boldsymbol{\pi} = \mathbf{p} - e\mathbf{A}/c$ and the gauge-field commutator $[\pi_i, \pi_j] = ie\hbar\epsilon_{ijk}B_k/c$ (so $\boldsymbol{\pi} \times \boldsymbol{\pi} = (ie\hbar/c)\mathbf{B}$):

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 - \frac{e\hbar}{c} \boldsymbol{\sigma} \cdot \mathbf{B} \quad (\text{III.10}). \quad (13)$$

[Wolfram-MCP: Simplify $[\boldsymbol{\pi}^2 + i \cdot (ie\hbar/c)\boldsymbol{\sigma} \cdot \mathbf{B}]$ returns the (III.10) RHS (DRQM I §III.10, verified).]

2.7 Eqs. (III.11)–(III.17) — spherical-coordinate algebra

Expand $(-i\hbar\boldsymbol{\sigma} \cdot \nabla V_0)(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})$ in spherical polar coordinates with the Coulomb scalar potential $V_0 = -e^2/r$ and the proton dipole vector potential $\mathbf{A} = (\boldsymbol{\mu}_p \times \mathbf{r})/r^3$. DRQM I records the chain across seven equations (III.11)–(III.17); the three load-bearing results that the chain delivers (DRQM I lines 329, 333, 338) are:

$$-i\hbar\nabla V_0 \cdot \mathbf{p} = \frac{e^2\hbar^2}{r^2} \frac{\partial}{\partial r}, \quad (\text{scalar piece; } \sim \text{velocity-radial coupling}) \quad (14)$$

$$\hbar\boldsymbol{\sigma} \cdot (\nabla V_0 \times \mathbf{p}) = -\frac{e^2\hbar}{r^3} \boldsymbol{\sigma} \cdot \mathbf{L}, \quad (\text{spin-orbit emergence; } \mathbf{L} = \mathbf{r} \times \mathbf{p}) \quad (15)$$

$$-\frac{e\hbar}{c} \boldsymbol{\sigma} \cdot (\nabla V_0 \times \mathbf{A}) = -\frac{e\hbar}{c} \cdot \frac{e^2}{r^4} 2\mu_p |\mathbf{s}_p| \sin\theta (\boldsymbol{\sigma} \cdot \mathbf{e}_\theta). \quad (\text{electron-nuclear cross term}) \quad (16)$$

(14) is the radial scalar piece (contributes to relativistic kinetic + Darwin once the Pauli/FW reduction is taken); (15) is the standard spin-orbit emergence $\boldsymbol{\sigma} \cdot \mathbf{L}/r^3$; (16) is the electron-proton dipole-dipole contribution with explicit angular dependence $\sin\theta (\boldsymbol{\sigma} \cdot \mathbf{e}_\theta)$.

Note on intermediate (III.11)–(III.16). DRQM I records the spherical-coordinate chain across seven numbered equations (III.11)–(III.17), but only the three load-bearing results above carry forward to (III.18)–(III.20); the intermediate equations (III.11)–(III.16) are the standard vector-calculus identities for $\nabla(1/r)$, $\nabla V_0 \cdot \mathbf{p}$, the cross-product $\nabla V_0 \times \mathbf{p}$, and the same with $\mathbf{p} \rightarrow \mathbf{A}$ (proton vector potential), each evaluated in $(\hat{r}, \hat{\theta}, \hat{\phi})$ basis. They appear in DRQM I lines 297–339; the verification doc summary records them as “standard spherical-coordinate vector calculus, algebra is mechanical.”

2.8 The three “new” terms (III.18)–(III.20)

Collecting (14)–(16) together with the Pauli-identity collapse (13) in the approximated eigenvalue equation (11), three contributions appear that are absent from the non-relativistic Schrödinger problem:

$$(\text{III.18}) \quad - \left[1 - \frac{4r_0}{2r + r_0} \right] \frac{e\hbar\boldsymbol{\sigma} \cdot \mathbf{B}}{2mc} \quad (\text{spin-magnetic, cutoff-modified}) \quad (17)$$

$$(\text{III.19}) \quad 2r_0\mu_p^2 |\mathbf{s}_p|^2 \left[1 - \frac{4e\hbar^2}{mc(2r + r_0)} \right] \frac{\sin^2\theta}{r^4} \quad (\text{square of nuclear-dipole field}) \quad (18)$$

$$(\text{III.20}) \quad -\frac{2e r_0 \hbar \mu_p |\mathbf{s}_p|}{mc(2r + r_0)} \left[1 + \frac{4r_0}{2r + r_0} \right] \frac{\sin\theta (\boldsymbol{\sigma} \cdot \mathbf{e}_\theta)}{r^3} \quad (\text{electron-nuclear cross term}) \quad (19)$$

(17) is the source of the modified g -factor; (18)–(19) are the hyperfine pieces.

2.9 The g -factor formula (III.21)–(III.22)

Reading off the Zeeman coefficient from (17) with $\mathbf{s} = \hbar\boldsymbol{\sigma}/2$:

$$H_a = 2 \left[1 - \frac{4r_0}{2r + r_0} \right] \mu_B \mathbf{s} \cdot \mathbf{B} \quad (\text{III.21}), \quad (20)$$

identifying the gyromagnetic ratio

$$\boxed{g_r(x) = 2 \left[1 - \frac{4}{2x + 1} \right], \quad x \equiv r/r_0} \quad (\text{III.22}). \quad (21)$$

Limit checks: $g_r(1/2) = -2$ (tree-Dirac), $g_r(x \rightarrow 0) = -6$. For muon and proton, Eq. (III.23) writes the analogue with r_0^μ, r_0^p but leaves the cutoff values r_μ, r_p as separate, numerically-unspecified parameters.

The closed-form algebraic inversion (§III.D-extension). Inverting (21) for $g_r(x) = -2(1 + a_e)$ with a_e the electron anomalous magnetic moment, and solving algebraically:

$$\boxed{\frac{r_e}{r_0} = \frac{2 - a_e}{2(2 + a_e)}} \quad (22)$$

With CODATA-full a_e^{expt} this reproduces the triangulated $r_e/r_0 = 0.499\,420\,509\,9128\dots$ to 3.45×10^{-13} (within the precision floor $\sigma_r = 2.5 \times 10^{-13}$); see §?? and §3. The honest framing of (22) is *reproduction-by-inheritance under hypothesis (ii)* (QED supplies $a_e(\alpha)$; the framework supplies the cutoff–anomaly identification + the algebraic inversion). An unconditional **Pass** requires hypothesis-(i); see issue #75.

2.10 From $g_r(r)$ to each observable

The campaign’s six g_s -dependent precision observables each have a prediction of the form

$$\mathcal{O}_i^{\text{framework}} = \left(\frac{g_s}{-2} \right)^{n_i} \cdot \mathcal{O}_i^{\text{textbook leading}} \quad n_i \in \{1, 2\} \quad (23)$$

with $g_s = g_r(r_e/r_0)$ from (21), evaluated at the triangulated cutoff. The n_i exponents are determined by the operator structure of each observable; the textbook leading term carries the rest of the physics (kinematics, multipole structure, Bethe logs).

2.10.1 Electron g_s ($n = 1$, directly from g_r)

Set $r = r_e$ in (21): the framework’s prediction is $g_s = g_r(r_e/r_0)$ directly. The triangulated r_e/r_0 is by construction the value at which $g_s = -2.00231930\dots$ matches CODATA-2018, so the match is exact by Pass-B fit construction (§??).

2.10.2 Hydrogen $2P_{3/2}$ – $2P_{1/2}$ fine structure ($n_{\text{FS}} = 1$)

The Sommerfeld–Dirac formula gives the leading fine-structure splitting

$$\Delta E_{\text{FS}}^{\text{leading}}(2P_{3/2} - 2P_{1/2}) = \frac{m_e c^2 (Z\alpha)^4}{32} (\text{angular factor}) \approx 10\,949 \text{ MHz at } Z = 1. \quad (24)$$

Spin–orbit is linear in g_s (it enters via the $\mathbf{L} \cdot \mathbf{S}$ Lande factor in the Pauli/FW reduction of the dual Dirac equation, BS §14.1), so $n_{\text{FS}} = 1$:

$$\Delta E_{\text{FS}}^{\text{framework}} = \frac{g_s}{-2} \Delta E_{\text{FS}}^{\text{leading}} \approx 1.00115965 \cdot 10\,949 \approx 10\,962 \text{ MHz}, \quad (25)$$

with the framework anomalous correction +12.7 MHz on top of the leading 10 949 MHz. Remaining ~ 7 MHz vs. measured 10 969.13(10) MHz is recoil + two-loop, at the Bethe-estimate precision floor (see §?? for the $Z^4 \text{ Li}^{2+}$ extension).

2.10.3 Hydrogen 1S hyperfine (21-cm line, $n_{\text{HFS}} = 1$)

The Fermi (1930) contact term gives the leading 1S hyperfine Hamiltonian

$$H_{\text{HF}} = \frac{8\pi}{3} g_p \mu_N g_s \mu_B \delta^3(\mathbf{r}) \mathbf{I} \cdot \mathbf{S}, \quad (26)$$

with $\langle 1S | \delta^3(\mathbf{r}) | 1S \rangle = 1/(\pi a_0^3)$, producing the splitting

$$\Delta E_{\text{HF}}^{\text{leading}}(1S_{1/2}) = \frac{4}{3} g_p \frac{m_e}{M_p} \alpha^4 m_e c^2 \approx 1\,418.4 \text{ MHz} \quad (\text{at } g_s = -2). \quad (27)$$

Hyperfine is *linear* in g_s ($n_{\text{HFS}} = 1$, since the contact term carries a single electron spin):

$$\Delta E_{\text{HF}}^{\text{framework}} = \frac{g_s}{-2} \Delta E_{\text{HF}}^{\text{leading}} \approx 1.00115965 \cdot 1\,418.4 \approx 1\,420.04 \text{ MHz}. \quad (28)$$

Residual vs. measured $1\,420.405\,751\,768(2)$ MHz is ~ 0.4 MHz — sub-leading QED (α/π + recoil + nuclear structure), r_e -independent, at the Bethe-estimate floor (BS-§22.1).

2.10.4 Lamb shift $2S_{1/2}$ – $2P_{1/2}$ — formulation-independent ($n = 0$)

The Lamb shift escapes the $(g_s/-2)^n$ family. Bethe’s (1947) mass-renormalisation argument writes the self-energy shift

$$\Delta E_n^{\text{SE}} = \frac{2\alpha}{3\pi m^2 c^2} \sum_m |\langle m | \mathbf{p} | n \rangle|^2 (E_m - E_n) \ln \frac{K}{|E_m - E_n|}, \quad (29)$$

where K is a non-relativistic UV cutoff. Under the proper-time formulation:

- matrix elements $|\langle m | \mathbf{p} | n \rangle|^2$ are formulation-independent (BS-§8);
- energy denominators $E_m - E_n$ are formulation-independent (BS-§4);
- the Bethe-log structure $\ln(K/|E_m - E_n|)$ is non-relativistic in origin and inherits the same UV cutoff;
- the mass-renormalisation subtraction is formulation-independent at the order Bethe needed.

Therefore $\Delta E_{2S}^{\text{SE, proper-time}} \approx 1\,040$ MHz is *identical* to the standard Bethe-estimate prediction at the precision the route delivers. Adding the vacuum-polarisation shift (-27 MHz at 2S, BS-§21) and recombining gives $\sim 1\,013$ MHz at the Bethe-estimate floor vs. measured $1\,057.845(9)$ MHz; the ~ 42 MHz residual is the gap to the full one-loop QED calculation, which the campaign does not produce (BS-§20). g_s does not enter the leading Bethe-log contribution ($\mathbf{p} \cdot \mathbf{p}$, not $\mathbf{s} \cdot \mathbf{B}$), so $n_{\text{Lamb}} = 0$ in (23).

2.10.5 Helium 3P_J fine structure ($n_{\text{He}} = 1$)

Two-electron spin-orbit + spin-spin recombination in the helium triplet 2P states produces the 3P_0 – 3P_1 interval. The leading Bethe–Salpeter (1957) treatment gives a $(Z\alpha)^4$ scale modulated by two-body recoil + correlation factors; spin-orbit is linear in g_s , so $n_{\text{He}} = 1$. Predicted framework value $\sim 29\,616.95$ MHz vs. measured $29\,616.952$ MHz at the kHz level, provisional residual after spin-spin/spin-orbit recombination (BS-§72).

2.10.6 Positronium ortho–para ($n_{\text{Ps}} = 2$)

The 1^3S_1 – 1^1S_0 ortho–para splitting in positronium (e^+e^-) involves a contact term *quadratic* in the electron magnetic moment (both leptons carry an anomalous moment that enters via the two-fermion spin-spin coupling), giving $n_{\text{Ps}} = 2$:

$$\Delta E_{\text{Ps}}^{\text{framework}} = \left(\frac{g_s}{-2} \right)^2 \cdot \Delta E_{\text{Ps}}^{\text{textbook leading}} \approx (1.00115965)^2 \cdot 203\,389 \text{ MHz}. \quad (30)$$

Per 10_CrossComparison.md §2, agreement is at Bethe precision; positronium-specific QED (the annihilation contribution) is provisional in the campaign (BS-§80).

2.10.7 Muonium hyperfine ($n_\mu = 1$)

Muonium (μ^+e^-) replaces the proton in (26) with an anti-muon. The Fermi-contact framework carries through with proton $\rightarrow$ anti-muon kinematics; spin-orbit / Fermi contact is linear in g_s , $n_\mu = 1$:

$$\Delta E_{\mu\text{HFS}}^{\text{framework}} \approx \frac{g_s}{-2} \cdot 4\,458.2 \text{ MHz} \approx 4\,463.4 \text{ MHz}, \quad (31)$$

vs. measured 4 463.302 776(51) MHz: ~ 0.1 MHz residual at the sub-leading-QED floor (BS-§80).

2.11 Summary of n_i exponents and observable map

Observable	n_i in $(g_s/-2)^{n_i}$	Operator structure
electron g_s	1 (direct)	$g_r(r_e/r_0)$ itself, Eq. (21)
H $2P_{3/2}-2P_{1/2}$	1	spin-orbit, $\mathbf{L}\cdot\mathbf{S}$
H 1S hyperfine	1	Fermi contact, $\delta^3(\mathbf{r}) \mathbf{I}\cdot\mathbf{S}$
He $^3P_0-^3P_1$	1	two-body spin-orbit (post-recombination)
positronium ortho-para	2	two-fermion spin-spin contact (both leptons)
muonium hyperfine	1	Fermi contact, $e^-\mu^+$
H Lamb shift $2S-2P$	0	formulation-independent self-energy (Bethe 1947)

What the chain establishes (and does not). Once (21) is in hand and the triangulated r_e/r_0 is fixed by Pass-B joint fit (§3), each g_s -dependent observable evaluates as $(g_s/-2)^{n_i} \times$ its textbook leading term — a single back-fit applied six times rather than six independent corroborations (10_CrossComparison.md §2; §7). The Lamb shift escapes the family entirely because the leading Bethe-log self-energy is g_s -symmetric. The chain’s content is the (III.22) cutoff-anomaly identification; the empirical $a_e^{\text{bound}}(Z\alpha)$ curve across the Z-scan is QED’s, per the Branch A vs. Branch C reading.

[for Tepper: This section transcribes the derivation chain you already laid out in DRQM I §§II–III, with every typeset numbered equation from (II.1) through (III.22) surfaced for fine-grain review per your 2026-05-30 instruction. The intermediate (III.11)–(III.16) spherical-coordinate-algebra equations are noted as “mechanical vector-calculus” per the verification doc’s own characterisation and not re-derived here; their three load-bearing outputs (14)–(16) are displayed. The per-observable n_i summary table is a synthesis move (drawn from the BS section docs but consolidated here for the first time in one place) and is endorsed per your 2026-05-30 sign-off.]

3 Z=1 empirical triangulation of r_e/r_0

Result.

$\frac{r_e}{r_0} = 0.4994205099128317 \pm 2.50 \times 10^{-13}$

(PR #62, joint best-fit across six observables).

Method (full derivation: §??). Per 10_CrossComparison.md §2, the six g_s -dependent observables each enter as $\text{anchor}_i \cdot (g_r(x)/g_s^{\text{meas}})^{n_i}$ with $n_i \in \{1, 2\}$. Two weighting passes are reported:

- **Pass A** (measurement- σ only): $r_{\text{opt}} = 0.4994061257148855$ at $\chi^2 \approx 2.97 \times 10^{16}$. Unphysical: pulls $g_e(r_{\text{opt}})$ off measured by 5.76×10^{-5} to compensate the hyperfine residual that the leading- $(g_s/-2)^n$ model cannot fit by any choice of r alone.

- **Pass B** (measurement- σ + framework-floor noise): $r_{\text{opt}} = \mathbf{0.4994205099128317}$ at $\chi^2 \approx 3.99958 \approx 4$ (the number of framework-floor residuals at $\sim$ one-floor). Matches branch (c) to 16 sig figs.

Pass B is the honest weighting. Stretched-fit flag: **none**.

Closed-form algebraic inversion (the §III.D-extension). Roadmapping/Equation_Verification/Dual §III.D-extension establishes the closed-form

$$\boxed{\frac{r_e}{r_0} = \frac{2 - a_e}{2(2 + a_e)}}$$

obtained by inverting $g_r(r_e/r_0) = -2(1 + a_e)$. With CODATA-full a_e^{expt} , this gives $r_e/r_0 = 0.499\,420\,509\,913\,18$, matching the triangulated optimum to 3.45×10^{-13} (within precision floor $\sigma_r = 2.5 \times 10^{-13}$).

Honest framing (verbatim from DRQM I §III.D-extension). This is *reproduction-by-inheritance under hypothesis (ii)*: QED supplies $a_e(\alpha)$; the framework supplies the (III.22) cutoff-anomaly identification and the algebraic inversion. It is **not** an independent dual-framework derivation of a_e . The §III.D Schwinger identification subsection (PR #70) characterises the same algebraic operation as the back-fit identity from the Candidate-3 perspective; both subsections together give the full picture.

Finding 2 verdict trajectory (verbatim from FINDINGS_for_author_review.md:209).

• (was) $\longrightarrow$ • characterised (#61) $\longrightarrow$ •/• at hypothesis-(ii) framework precision (#64)

Unconditional • requires hypothesis-(i) re-derivation (proper-time photon propagator + bound-state propagator in the (II.3) Pauli kernel + mass-renormalisation prescription at the cutoff), tracked in **issue #75**.

Hypothesis split (the load-bearing question).

- (i) the dual-framework proper-time one-loop vertex (with the dual photon propagator and bound-state structure) is computed *independently* and shown to reproduce $a_e^{(1)} = \alpha/(2\pi)$;
- (ii) the dual-framework one-loop vertex correction *reproduces* the standard-QED Schwinger anomaly identically by inheritance, with all dual structure absorbed into the matter-side (II.3) kernel which at one-loop is formulation-independent.

The closed-form inversion + CODATA- a_e chain delivers the triangulated value under either reading; only (i) constitutes a derivation *within* the framework.

Variational / operator-coefficient route (Candidate 2). Route Z's framework-internal closure (closure 7c on the spin-magnetic operator coefficient, tree-level target $g = -2$) yields $r_e/r_0 = 1/2$ exactly, matching FoundationsII-Classical §2.2's critical-point identification. The discrepancy with the triangulated value 0.4994205099128317 is 5.79×10^{-4} — exactly the size of the Schwinger one-loop QED correction $\alpha/(2\pi) \approx 1.16 \times 10^{-3}/2 \approx 5.8 \times 10^{-4}$. The Schwinger 1-loop refinement reproduces the triangulated value to $\sim 10^{-7}$ (Karplus-Kroll-level), but requires QED input *external* to the framework. Full derivation: §?? and §??.

[for Tepper: (1) Does the Pass B weighting match your intent? (2) Is “reproduction-by-inheritance under hypothesis (ii)” the framework's intended position for the closed-form result, or is hypothesis (i) (issue #75) required? (3) Does framing the triangulated value as a *refinement* of the published initial-value r_e — rather than as a result the published derivation must reproduce — match your reading?]

4 Z-scan empirical refinement

Headline result (PR #87, full derivation: §??). The multi- Z scan across $Z = 2, 6, 8, 14, 50$ supplies the cross- Z comparison the Li^{2+} branches (§??-??) could not run.

$$\chi_{(Z-i) \text{ universal cutoff}}^2 = 1.0282 \times 10^{16} \implies \text{OUTCOME A DECISIVELY REJECTED.}$$

Ion set + measurement provenance. $^3\text{He}^+$ (Schneider 2022 *Nature* 606, 878); $^{12}\text{C}^{5+}$ (Sturm 2014 *Nature* 506, 467); $^{16}\text{O}^{7+}$ (Verdú 2004 *PRL* 92, 093002); $^{28}\text{Si}^{13+}$ (Sturm 2011 *PRL* 107, 023002); $^{118}\text{Sn}^{49+}$ (Morgner 2023 *Nature* 622, 53). Combined absolute 1σ on g : ranges 10^{-11} to 10^{-9} .

Per- Z back-fit cutoff $x^{(Z)} = r_e^{(Z)}/r_0$. Inverting $g_r(x) = -2(1+a)$ at each measured g :

$$x^{(Z)} = \frac{2 - a_{\text{bound}}^{(Z)}}{2(2 + a_{\text{bound}}^{(Z)})}.$$

Numerical results (Wolfram-MCP 2026-05-27, STATE.md iter 11):

Z	a_{bound}	$x^{(Z)}$
1	a_e^{CODATA}	0.4994205099128317 ($= x_{\text{univ}}$)
2	+0.00108870790	0.49945594221
6	+0.00052079510	0.49973967024
8	+0.00002351270	0.49998824379
14	−0.00232552065	0.50116411391
50	−0.04471897050	0.52287086604

Z-scaling fit (reading (Z-ii)). Fitting $x^{(Z)}$ vs. $(Z\alpha)^2$:

$$\text{linear: } x = 0.499\,383\,5903 + 0.176\,393\,13 (Z\alpha)^2$$

$$\text{quadratic: } x = 0.499\,420\,6084 + 0.166\,275\,68 (Z\alpha)^2 + 0.074\,154\,07 (Z\alpha)^4$$

- Quadratic intercept $c_0 = 0.4994206084$ matches $x_{\text{univ}} = 0.4994205099$ to 9.8×10^{-8} .
- Quadratic slope $c_2 = 0.166276$ matches QED’s $1/6 = 0.166667$ to 2.3×10^{-3} .

The clean $(Z\alpha)^2$ form is a property of QED’s bound-state $g(Z\alpha) = 2[1 - (1/3)(Z\alpha)^2 - \dots]$, *inherited* by the back-fit — not derived from the framework apparatus.

Outcome C (per PR #87 conclusion, verbatim).

- Outcome A (single universal cutoff) excluded at $\chi^2 \sim 10^{16}$ over 5 ions.
- The (Z-ii) back-fit $x^{(Z)}$ is well-described by $x = c_0 + c_2(Z\alpha)^2 + c_4(Z\alpha)^4$ with $c_0 = x_{\text{univ}}$ recovered as $Z \rightarrow 0$ to 10^{-7} , and $c_2 \approx 1/6$ — the QED bound-state leading coefficient.
- The clean $(Z\alpha)^2$ form is QED-inherited, not framework-derived. Therefore **Outcome C: Z-specific cutoff through QED-inheritance**, extending PR #70’s lepton-axis verdict (“particle-specific through a_ℓ ”) to the Z -axis (“ Z -specific through $a_e^{\text{bound}}(Z\alpha)$ ”).

Reading. The Z-scan does not contradict the algebra of PRs #84/#85/#86 (the (III.22) formula indeed has no explicit Z). What it shows is that the framework’s apparatus cannot *derive* a Z -dependent inheritance curve — the framework supplies $g_r(x)$ and leaves each state’s cutoff free (per (III.23)); the per- Z back-fit lands on the QED bound-state curve because that’s the empirical content of $a_e^{\text{bound}}(Z\alpha)$, supplied to the inversion as input. The Branch A vs. Branch C tension reduces to which reading of (III.22)/(III.23) the senior author intends — §5.

[for Tepper: The PR #87 numerical findings are reproduced verbatim from the notebook. The interpretive move from “Outcome A excluded” to “Outcome C = QED-inheritance” is substantive AI: please confirm whether the Z-scan empirical exclusion of Outcome A and the identification of the back-fit slope with the QED 1/6 matches your reading of (III.23).]

5 Branch A vs. Branch C — the tension presented without endorsement

Four open PRs against `main` make verdicts about r_e/r_0 that need reconciliation, and the reconciliation is exactly what this packet asks the senior author to provide.

Branch A (Z-universal cutoff) — PRs #84, #85, #86.

- Reads (III.22)’s closed form $r_e/r_0 = (2 - a_e)/(2(2 + a_e))$ as Z -independent: a_e is the free-electron anomaly, $r_0 = e^2/(m_e c^2)$ depends only on electron mass.
- “The radiating particle is the same electron at every Z , so r_e/r_0 is identical at $Z = 1$ and $Z = 3$.” (`r_e_Li2plus_joint_fit.wl`, gating result.)
- Verdict: **Branch A by construction**; Branch B (framework-derivable Z -scaling) ruled out structurally.
- Empirical context: three of four Li^{2+} comparisons are **Blocked** (mis-provenanced brief measurements — §6), so PRs #84/#85/#86 cannot compute cross- Z residuals.

Branch C (per- Z QED inheritance) — PR #87.

- Supplies the cross- Z residual table at $Z = 2, 6, 8, 14, 50$ using valid measurements.
- Tests (Z-i) universal cutoff: $\chi^2 = 1.0 \times 10^{16} \Rightarrow$ **Outcome A decisively rejected**.
- Per- Z back-fit $x^{(Z)}$ follows $x = c_0 + c_2(Z\alpha)^2 + c_4(Z\alpha)^4$ with $c_0 = x_{\text{univ}}$, $c_2 \approx 1/6$ (QED’s bound-state coefficient).
- Verdict: **Outcome C** — Z -specific cutoff through QED-inheritance. “The verdict shape is Branch C, not Branch A.” (PR #87 body, verbatim.)

The non-contradiction reading (substantive AI synthesis). The two verdicts can both be honest readings of different things. The algebra of (III.22), taken literally, has no Z in the coefficients — Branch A is correct as an algebraic statement *about the formula*. The empirical data across Z rejects fixing a single x value across all Z — Branch C is correct as an empirical statement *about the fitted cutoff in nature*.

The synthesis: *the framework’s apparatus supplies the (III.22) formula $g_r(x)$ and leaves each state’s cutoff free per (III.23); the apparatus does not internally derive the binding coefficient $-(Z\alpha)^2/3$. So the framework, as published, is consistent with Branch A (its formula has no Z) and the empirical $r_e^{(Z)}/r_0$ inherits QED’s bound-state structure when fit ion-by-ion. The framework’s content here is the (III.22) identification of the cutoff with the anomaly via the $g_r(x)$ formula, not a derived prediction of $a_e^{\text{bound}}(Z\alpha)$.*

Implication for the campaign-closing statement. The reading above does not change the honest-scope statement (zero of 28 results constitute independent corroboration distinct from textbook QM); it sharpens *why*: the (III.22) cutoff-anomaly identification is the framework’s contribution; the empirical $a_e^{\text{bound}}(Z\alpha)$ curve is QED’s. Both readings agree that the framework does not internally derive the binding coefficient. The disagreement is about how to label the resulting verdict — “Branch A by construction” (algebraic, Li^{2+} -branches) vs. “Branch C QED-inheritance” (empirical, Z-scan). The label choice does not change any prediction or honest-scope statement; it changes what `10_CrossComparison.md` §1 should say going forward about the r_e resolution’s *character*.

Verdict question for the senior author. Both verdicts on `FINDINGS_for_author_review.md` Finding 2 must land on `main` consistently, and the sequencing of merges across the four BS PRs depends on which label the senior author endorses:

- **If Branch A by construction is preferred:** PR #87’s $\chi^2 = 10^{16}$ result needs to be re-framed in the merged text as “the per- Z back-fit cutoff curve is QED-inherited; the (III.22) algebra produces the universal $r_e/r_0 = x_{\text{univ}}$ as the $Z \rightarrow 0$ limit, and the inheritance curve is empirical content outside the framework’s algebraic apparatus.”
- **If Branch C QED-inheritance is preferred:** PRs #84/#85/#86’s “Branch A by construction” verdicts need to be qualified as “algebra-correct but not the framework-final verdict; the empirical inheritance per (III.23) is the operative reading.”

[for **Tepper**: This is the central reconciliation question of the packet. Which label is the framework’s intended verdict, and how should `10_CrossComparison.md` §1 read going forward?]

This packet takes no position. Both readings are presented above as honestly as we know how to. The adjudication belongs to the senior author and is exactly what §10 asks for.

6 Mis-provenance audit log (Li^{2+} brief)

Three of four Li^{2+} “measurements” in the original brief for issue #78 turned out to be mis-provenanced. The audit log below is transcribed from the bodies of PRs #84, #85, and #86, and is the load-bearing reason those PRs’ *cross-Z residual* computations are blocked (predictions are not blocked — they are computed in §??-??).

Mis-provenance #1 (Observable #1, bound g -factor of ${}^7\text{Li}^{2+}$).

- **Brief citation:** “Sturm 2014 *Nature* 506, 467” for $g_{\text{bound}}({}^7\text{Li}^{2+}) = 2.0000251707$.
- **What that paper actually measured:** the ${}^{12}\text{C}^{5+}$ electron mass (i.e. $Z = 6$, not $Z = 3$).
- **Cross-check:** the cited value 2.0000251707 Wolfram-inverts to $Z \approx 8.04$, not $Z = 3$.
- **Framework prediction at $Z = 3$:** $g_{\text{bound}} \approx 2.00200$ (§??).

Mis-provenance #2 (Observable #3, $2P_{3/2}-2P_{1/2}$ FS in ${}^7\text{Li}^{2+}$).

- **Brief citation:** $\sim 7,367$ MHz (~ 7.4 GHz).
- **What that value actually is:** helium-like Li^+ ($1s2s-1s2p$, two-electron) fine structure, *not* hydrogenic Li^{2+} .
- **Cross-check:** hydrogenic Li^{2+} FS scales as $Z^4 \cdot 10,949$ MHz = 886,892 MHz $\approx$ 887 GHz (§??), ~ 2 orders of magnitude above the brief’s value.
- **Riis et al. 1994 attribution** (also in the brief) is He-like Li^+ 2-electron Lamb shift, not hydrogenic Li^{2+} either.

- **Framework prediction:** 887,920 MHz $\approx$ 887.92 GHz (Z^4 -scaled leading Dirac + anomalous offset +1028 MHz).

Mis-provenance #3 (Observable #4, ^7Li 1s hyperfine).

- **Brief citation:** “Beckmann 1974 *Z. Phys.* 270, 173” for ~ 12.7 GHz.
- **What that paper actually is:** a nuclear-moment measurement, *not* a Li^{2+} hyperfine measurement.
- **Cross-check:** the framework prediction (29.85 GHz, $^3\text{He}^+$ -method-validated to 0.1%) is off the brief’s value by $2.35\times$.
- **Framework prediction:** 29 846 MHz $\approx$ 29.85 GHz (§??). No direct experimental $^7\text{Li}^{2+}$ HFS was located in targeted search; the QED-theory comparator is Pachucki, Patkos, Yerokhin 2023, arXiv:2309.00436, Table VII (which uses experimental Li^+ HFS as input).

Mis-provenance #4 (none — Observable #2 is valid).

- **Brief citation:** Schiffer 1995 *PRL* 74, 2188 for the $2S_{1/2}-2P_{1/2}$ Lamb shift in $^7\text{Li}^{2+}$ at 62,765(21) MHz.
- **Status:** valid hydrogenic Li^{2+} Lamb shift measurement.
- **Framework prediction:** $\sim 61.1\text{--}62.7$ GHz Bethe-estimate (§??); agreement at framework precision, r_e -independent (weak discriminator).

Implication for the Li^{2+} verdict. Because #1, #3, #4 all carry valid framework predictions but lack valid measurements, the four-observable joint χ^2 at $Z = 3$ is not computable. The Li^{2+} branches’ “Branch A by construction” verdict is reached *in the absence of valid cross- Z residuals*. PR #87’s Z -scan supplies the cross- Z residual table the Li^{2+} branches could not run (§4).

[for **Tepper:** The audit log above is transcribed verbatim from PR bodies #84/#85/#86. Two questions: (a) do you have access to valid published $^7\text{Li}^{2+}$ measurements for #1, #3, #4 that the targeted searches in this campaign missed? (b) is the Schiffer 1995 attribution for the Li^{2+} Lamb shift the citation you would carry forward in the manuscript?]

7 Six-observable cross-comparison + NIST-MCP provenance audit

The campaign’s load-bearing quantitative comparison is the 10_CrossComparison.md §2 six-observable table. The cross-check column was added in PR #91 (issue #90) using the nist MCP server [1]; the column is a per-observable provenance audit, not an independent corroboration.

Observable	Framework prediction at triangulated r_e	Measurement	Residual	Source of residual	NIST cross-check (#90)	MCP
Electron g_s (Finding 2 itself)	−2.00231930	−2.00231930... (CODATA)	0	matches by construction	−2.00231930436256(35) CODATA 2018, get_constant	
Hydrogen $2P_{3/2}-2P_{1/2}$ (PR C)	$\sim 10\,962$ MHz	10 969.13(10) MHz	~ 7 MHz	Bethe-estimate floor (recoil 2-loop) +	10 969.049(15) MHz ASD levels, theoretical/Ritz , get_levels	

Observable	Framework prediction at triangulated r_e	Measurement	Residual	Source of residual	NIST cross-check (#90)	MCP
Hydrogen 1S hyperfine (PR F)	$\sim 1\,420.04$ MHz	$1\,420.405\,751\,768(2)$ MHz	~ 0.4 MHz	sub-leading QED (α/π + recoil + nuclear-structure)	$1\,420.405\,751\,768(2)$ MHz curated, list_dirac_targets	
Helium $^3P_0-^3P_1$ (PR I)	$\sim 29\,616.95$ MHz	$29\,616.952$ MHz	$\sim$ kHz	full QED residual (provisional — spin-spin/spin-orbit recombination not derived in PR I)	$29\,616.952$ MHz curated, list_dirac_targets	
Positronium ortho-para (PR I)	$\sim 203\,389$ MHz	$203\,389(2)$ MHz	matches at Bethe precision	positronium-specific QED (provisional — absorbing correction not derived in PR I)	$203\,389(2)$ MHz curated, list_dirac_targets	
Muonium hyperfine (PR I)	$\sim 4\,463.4$ MHz	$4\,463.302\,776(51)$ MHz	~ 0.1 MHz	sub-leading QED	$4\,463.302\,776(51)$ MHz curated, list_dirac_targets	

Reading the cross-check column: the electron g_s entry is the only one carrying an authoritative independent source (CODATA-2018). The hydrogen 2P fine-structure entry is ASD-theoretical (Ritz level difference), not a measurement; it differs from the measured value by ~ 0.08 MHz. The remaining four entries are curated transcriptions of the measurement column from the `list_dirac_targets` registry (the `nist` MCP server, PR #91 / issue #90), with provenance metadata only — they restate, not corroborate.

Reading the table honestly.

- Only the electron g_s row carries an authoritative independent source: CODATA-2018 via `scipy.constants` (`get_constant`). The MCP-retrieved value matches the measurement column to 16 sig figs.
- The H $2P_{3/2}-2P_{1/2}$ row’s MCP cross-check is the ASD *theoretical*/Ritz level difference at $\approx 10,969.05$ MHz, *not* an independent measurement. It differs from the measured $10,969.13(10)$ MHz by ~ 0.08 MHz. Do not read this as corroborating the framework.
- The H 1S hyperfine, He $^3P_0-^3P_1$, positronium ortho-para, and muonium hyperfine rows are served by the curated `list_dirac_targets` registry — they restate the measurement column with provenance metadata, but are not independent of it.

Pattern is consistent — and consistent by construction. All six observables share the same $(g_s/-2)^n$ scaling, and the triangulated r_e is by construction the joint-best-fit value across all six. The “six observables consistent” pattern is therefore a single-factor pattern (single-parameter joint fit) repeated six times, not six independent corroborations.

Honest reading (verbatim from 10_CrossComparison.md §2).

If the dual-theory framework is taken to require a single-parameter r_e calibrated against g_s , then the triangulated $r_e \approx 0.4994205099\,r_0$ is the joint-best-fit value across the g_s -dependent precision spectroscopy of hydrogen, helium, positronium,

and muonium. This is a self-consistency statement about parameter-fitting within the framework, not an independent experimental discrimination of the dual-theory framework from standard QED.

[for **Tepper**: Is the NIST-MCP cross-check column correctly framed as a retrieval / provenance audit (one independent CODATA source, one ASD-theoretical cross-check, four curated transcriptions), and not a corroboration of the framework?]

8 Open theoretical questions — issues #54 and #75

Two open theoretical threads bear directly on Finding 2’s unconditional ● verdict.

Issue #54 — first-principles derivation of r_e from the dual-Dirac renormalisation prescription. The empirical resolution of Finding 2 is settled at the $Z=1$ triangulated $r_e \approx 0.4994205099 r_0$ per PR #62. The theoretical resolution — whether a first-principles derivation from the dual-Dirac renormalisation prescription reproduces this cutoff — remains open.

Under the two readings of (III.22)/(III.23) the packet surfaces:

- Under **Branch A by construction**, the framework derivation would need to produce the value $r_e/r_0 = (2 - a_e)/(2(2 + a_e))$ with a_e supplied as input (i.e. derive the formula, leave the anomaly as a parameter).
- Under **Branch C QED-inheritance**, the framework derivation would need to produce the Z -curve $x^{(Z)} = c_0 + c_2(Z\alpha)^2 + c_4(Z\alpha)^4 + \dots$ with $c_2 \approx 1/6$ as a derived coefficient (not inherited from QED).

The two readings make different demands on what #54’s derivation, if pursued, would need to deliver.

Issue #75 — hypothesis-(i) proper-time photon-propagator re-derivation. The closed-form $r_e/r_0 = (2 - a_e)/(2(2 + a_e))$ reproduces the triangulated value under *hypothesis (ii)* (QED supplies $a_e(\alpha)$; the framework supplies the cutoff-anomaly identification and the algebraic inversion). Unconditional ● requires *hypothesis (i)*: the proper-time photon propagator + bound-state propagator in the (II.3) Pauli kernel + mass-renormalisation prescription at the cutoff, computed within the framework to reproduce $a_e^{(1)} = \alpha/(2\pi)$ independently.

The three framework specifications gating the hypothesis-(i) calculation are the load-bearing pieces of #75. Until they are specified, the verdict on Finding 2 stays at ●/● (at hypothesis-(ii) framework precision).

Outcome-matrix readings (per master #67). The packet’s findings line up with the master outcome matrix:

- **Outcome A** (cutoff produces triangulated value directly): excluded at $\chi^2 \sim 10^{16}$ over 5 ions by the Z -scan (PR #87), unless “triangulated value” is read as the $Z=1$ intercept only.
- **Outcome B** (framework-derivable Z -scaling): ruled out structurally by all four BS PRs (the algebra of (III.22) has no Z).
- **Outcome C** (per-state QED-inheritance, framework leaves cutoff free per (III.23)): the consistent reading once Outcomes A and B are excluded. This is the reading the variational notebook (§??) reaches as “Outcome C: framework-internal first-principles $r_e/r_0 = 1/2$ exactly, $\alpha/(2\pi)$ shift is QED-external,” and that the Z -scan (§??) confirms empirically.
- **Outcome D** (framework derivation produces yet a different value): excluded by the algebraic identification in §?? unless the framework’s (III.22) formula itself is revised.

[for **Tepper**: (1) Is the outcome-matrix reading above — A excluded by Z-scan, B excluded structurally, C the consistent residual, D unlikely without revising (III.22) — consistent with your reading of the campaign? (2) Does the unconditional-✓ path through issue #75 remain the right downstream sequencing for the verdict, or has your reading shifted?]

9 AI-use disclosure (Crocco §5)

Per Crocco, Rasdi & Garavan (2026), *Responsible AI in Non-Empirical Research*, and the repo’s implementation in `Roadmapping/Tooling/CROCCO_COMPLIANCE.md`, this packet was generated with explicit AI involvement that is disclosed below in two categories.

Pragmatic AI uses (formatting, transcription, retrieval).

- Wolfram-MCP symbolic verification of each load-bearing equation in the source `.wl` notebooks. Inline Wolfram-MCP confirmation markers throughout the equations sections record the date of each cell’s run. The `.wl` notebooks themselves are committed under `Roadmapping/Mathematica_Notebooks/Quantum_Mechanics/`.
- CODATA, NIST ASD, and PDG data retrieval via the `nist` MCP server (issue #90, PR #91, [1]) for the cross-comparison column.
- Bibliography stub scaffolding via `Roadmapping/History/Bibliography/scaffold_bib_note.py` and `build_bibtex.py`.
- Translation of source `.wl` notebook cells to LaTeX equation blocks (no derivation content changed; faithful transcription).

Substantive AI uses (synthesis, framing, prose drafting).

- Drafting of the packet’s prose under explicit human direction. Prompt-of-record: `prompts/implementation`.
- Selection of which load-bearing equations to surface in each `equations/ .tex` block (the cells judged most relevant to the senior author’s verdict question, not every cell of every notebook).
- Synthesis of the derivation-chain section (Dual Dirac Equation $\rightarrow$ each observable’s prediction). The equations are transcribed verbatim from DRQM I §§II–III and the BS section docs. After the senior author’s 2026-05-30 instruction, every typeset numbered DRQM I equation from (II.1) through (III.22) is now surfaced for fine-grain review; the only equation subset *not* re-derived is (III.11)–(III.16) (the spherical-coordinate vector calculus that the verification doc itself records as “mechanical algebra”), whose three load-bearing outputs are displayed. The remaining substantive synthesis move is (b) the per-observable n_i summary table consolidating BS section docs in one place, endorsed 2026-05-30.
- The Branch A vs. Branch C synthesis in §5 (“the algebra is correct as a statement about the formula; the empirical residual rejects Outcome A; Outcome C is the consistent residual”). This is a substantive interpretive move and is explicitly flagged.
- The outcome-matrix mapping in §8 (A excluded by Z-scan, B excluded structurally, C the consistent residual, D unlikely without revising (III.22)).
- Selection of the six closing questions in §10 from the larger pool of open questions inherited from the four BS PR bodies, the master #67 outcome-matrix issue, the FINDINGS Finding 2 verdict trajectory, and the open theoretical issues (#54, #75).

Substantive AI not present in this packet.

- No new physics derivations are introduced. Every equation in the `equations/` sections is transcribed from a committed `.wl` notebook with its Wolfram-MCP confirmation preserved.
- No verdict on Branch A vs. Branch C is announced by the packet; adjudication is deferred to the senior author per the closing question list.
- No abstract claim is made that the packet itself adjudicates the r_e/r_0 story. The packet’s purpose is to surface the already-recorded findings in the form (PDF, LaTeX equations) the senior author actually reads.

Cross-references to repo Crocco infrastructure.

- Compliance framework: `Roadmapping/Tooling/CROCCO_COMPLIANCE.md`.
- Substantive prompts directory: `Roadmapping/Tooling/_prompts/`.
- Packet-specific prompt-of-record: `author_review_packet/prompts/implementation_prompt.md`.

What the senior author should read against this disclosure. The packet’s prose is AI-drafted; the equations are AI-transcribed from human-authored `.wl` notebooks; the verdicts and the honest-scope statements are human-authored and preserved verbatim. Any substantive interpretive move (Branch A vs. Branch C synthesis, outcome-matrix mapping, closing-question selection) is flagged inline with `[ai-drafted: ...]` and surfaced explicitly in this section for transparent review.

[for Tepper: Please confirm the Crocco-compliance disclosure above accurately covers the AI involvement in this packet, and that the substantive interpretive moves listed are correctly distinguished from the pragmatic transcription tasks. Any addition or correction should land here before journal submission.]

10 Questions for Tepper

- Q1. **Z-scan dataset validity (PR #87).** Are the five Z-scan ions and their measured bound-electron g -factors valid as a cross- Z comparison set? $^3\text{He}^+$ Schneider 2022 *Nature* 606, 878; $^{12}\text{C}^{5+}$ Sturm 2014 *Nature* 506, 467; $^{16}\text{O}^{7+}$ Verdú 2004 *PRL* 92, 093002; $^{28}\text{Si}^{13+}$ Sturm 2013 *PRL* 110, 263002 (note: PR cites Sturm 2011 *PRL* 107, 023002 for $^{28}\text{Si}^{13+}$ — **[for Tepper: please confirm which Sturm paper is the right citation]**); $^{118}\text{Sn}^{49+}$ Morgner 2023 *Nature* 622, 53. If yes, the Z-scan dataset is sound and the (Z-i) universal-cutoff hypothesis is empirically dead at $\chi^2 \approx 10^{16}$.
- Q2. **Reading of Eq. (III.23).** PR #87 reads (III.22) as Z -independent and (III.23) as state-by-state freedom (the cutoff is left free per state, no algorithmic Z -coupling). PRs #84/#85/#86 read the same equations as fixing Branch A (Z -universal cutoff by construction). *What is the framework’s intended position on the cutoff freedom in (III.23)?*
- Q3. **Branch C as “QED-inheritance” — acceptable framework position?** The Z-scan’s empirical $x^{(Z)}$ curve follows $x = c_0 + c_2(Z\alpha)^2 + c_4(Z\alpha)^4$ with $c_2 \approx 1/6$, exactly QED’s bound-state coefficient. The framework’s apparatus does not derive $1/6$; the back-fit inherits it from the empirical $a_e^{\text{bound}}(Z\alpha)$ supplied to the inversion. *Is “the framework borrows bound-state QED’s $a_e(Z\alpha)$ rather than deriving the binding correction itself” an acceptable framework position, or does it amount to admitting the framework does not deliver a distinct Z -curve?*
- Q4. **Finding 2 verdict status.** `FINDINGS_for_author_review.md` Finding 2 currently sits at / at hypothesis-(ii) framework precision (per the #64 update). Unconditional  awaits

the hypothesis-(i) re-derivation in issue #75. *If Branch A is empirically dead (per PR #87) and the consistent reading is Branch C QED-inheritance, does Finding 2's verdict need to be downgraded from ●/● back to ● flagged until issue #54 resolves under the Branch C reading? Or does the Branch C reading already constitute the resolution?*

- Q5. **Mis-provenanced Li^{2+} measurements.** Three of four Li^{2+} measurements in the original brief were mis-provenanced (§6): brief's $g_{\text{bound}}(^7\text{Li}^{2+}) = 2.0000251707$ is a $Z \sim 8$ value (Sturm 2014 measured $^{12}\text{C}^{5+}$); brief's 7367 MHz 2P FS is helium-like Li^+ (2-electron), not hydrogenic Li^{2+} (whose value scales as $Z^4 \cdot 10,949 \approx 887$ GHz); brief's ~ 12.7 GHz HFS cites Beckmann 1974, a nuclear-moment paper not a Li^{2+} HFS measurement. *Do you know of valid published hydrogenic $^7\text{Li}^{2+}$ measurements for any of these three (PRs #84 and #86 logged that none were located in targeted search)?*
- Q6. **Triangulated $r_e/r_0 = 0.4994205099128317$.** Per your 2026-05-25 author guidance, the as-published r_e in DRQM I Eq. (III.22) was a uni-observable numerical search against g_s , not a derived quantity. PR #62's triangulation refines that to the joint best-fit across all six g_s -dependent observables at Pass B weighting, matching branch (c) to 16 sig figs. *Does the triangulated joint-fit value at $Z = 1$ stand as a calibration, even if Branch A is empirically dead (per PR #87) and the value is not Z -universal?*
- Q7. **Closed-form $r_e/r_0 = (2 - a_e)/(2(2 + a_e))$ — hypothesis (i) or (ii)?** The closed-form algebraic inversion (DRQM I §III.D-extension) matches the triangulated value to 3.45×10^{-13} under hypothesis (ii) (QED supplies $a_e(\alpha)$; framework supplies the cutoff-anomaly identification + the inversion). Under hypothesis (i) (issue #75), the proper-time one-loop vertex would need to be computed within the framework to reproduce $\alpha/(2\pi)$ independently. *Is hypothesis (ii) the framework's intended position for the closed-form result, or is hypothesis (i) (issue #75) required to constitute a derivation within the framework?*

Please leave your responses in line in this document, or add a comment on issue #93 with the answers. The four BS PRs (#84/#85/#86/#87) will be merged in the order your answers direct; 10_CrossComparison.md §1 will be updated to match.

References

- [1] Trey Morris and Claude Opus 4.7. nist_mcp: NIST ASD + CODATA MCP server (issue #90). GitHub PR #91, temoTxt/PyPhysics, May 2026.